

Professional Assessment

When I started at SNHU in October of 2022 I honestly didn't know a thing about computer science, I actually wasn't even interested in computer science until I took the Intro to Programming course as an elective and decided to completely change course. Over these last few years I've learned so much and have been able to develop my technical, analytical, and professional skills through all of the course work I've had to do. I've gotten experience in a wide range of things in computer science including software engineering, databases, algorithms, software security, software design, and user experience. The purpose of this ePortfolio is to demonstrate my growth throughout these last four years, by showing where I've come from and what I can accomplish today I can demonstrate my ability to create polished, professional-level work.

During my time at SNHU I learned how important it is to work and engage with others. Through discussion boards, peer reviews, and project feedback from professors I was able to see how different perspectives could be a positive contribution to my work. In classes like CS250 we learned about different Agile concepts and their importance to creating a cohesive collaborative environment during the software development lifecycle. We also did collaborative work where we had a team of students and each of us played a different role in a development team. In CS255 we did a project called DriverPass where we needed to interact with stakeholders in order to understand their expectations and business needs, and then translate them into technical documentation and design. This taught me that understanding a stakeholder's expectations becomes critical to delivering an effective product. These two experiences helped me understand that successful development relies on good communication as much as it does technical ability.

My software engineering skills have grown dramatically since starting this program. When I first started, my understanding of software engineering concepts was very limited, but now I can confidently say I've learned object-oriented programming principles and software architecture and maintainable design. This is evident in my software design and engineering artifact where I enhanced my Raspberry Pi thermostat application. The original met all the requirements and worked perfectly fine for the assignment, yet it wasn't even close to something you could call professional. In that enhancement I focused on object-oriented design, improved maintainability, readable code, exception handling, and input validation. All of these different aspects work together to show I have gained the skills to improve existing software and implement professional development practices.

Algorithms and data structures is another area where I've grown. I started out being very intimidated by the jargon, algorithms and data structures were terms I had never really heard before yet classes like CS260 introduced them to me and turns out they're not as scary as I first thought. In that class we learned how different algorithms can impact a program's performance and how scalable it is. We also got the opportunity to learn new ideas like searching, traversal, and recursion, additionally we learned about binary search trees which is what I ended up doing my enhancement on. The artifact for this enhancement was a course catalog that was powered by a binary search tree, in order to enhance it I improved the search functionality, added better validation, and made some other tweaks to the user experience so the program just feels better to use. By making these changes I gained a deeper understanding of the way data structures can directly affect the usability of software.

Databases were an area I initially struggled with because it felt different to traditional programming, yet as I've become more familiar with it I've started to enjoy it because it's the best way to store loads of data. Through a couple of different courses like CS340 we learned how to properly design a database and manage all of its data, we also got a lot of experience in the creation, use, and maintenance of databases. The artifact I chose for my database enhancement was the Animal Shelter database project we did in CS340, this enhancement really focused on making the database easier to use and improving the user experience. By adding better data management, focusing on application usability, and integrating with database operations I was able to gain insight into how important it is to have reliable, efficient, and user friendly experiences when developing a database driven application.

Software security was something that came later in the program, and something that threw me for a bit of a loop because of how different the mindset is from creating simple scripts. In our CS305 course we focused on software security, we learned about new ideas like input validation, error handling, and secure coding practices. These were all very different concepts to anything I had learned before but actually ended up making a lot of sense once I started to understand them better. I demonstrated how important these concepts are by integrating them into my enhancements, I noticed while going over some of my old programs that there was almost no consideration for software security so it ended up becoming a big focus of mine. It also became apparent to me while enhancing these projects that it's incredibly important to consider security all throughout the development process rather than add it in at the end.

This portfolio tells a story about my growth across multiple areas of computer science. The Raspberry Pi thermostat demonstrates my ability to design and develop effective software, create reliable architecture, and create maintainable code. The Course Catalog project shows my ability to work with data structures, effectively use algorithms, and solve complex problems. The Animal Shelter Database also demonstrates strong database management, efficient data organization, and strong consideration for user interaction. Together all of these artifacts and their enhancements help represent my skills and how they've developed through my time as a student in this program.

As I've created this ePortfolio it's become abundantly clear to me that I've seen significant growth since I initially joined the program in October 2022, I had basically no exposure to computer science then yet now I can develop professional level software. This has helped me tremendously in my confidence when it comes to approaching projects, I just have more faith that I can figure it out than I might've had 4 years ago. As my time through the program went on I have gained a stronger understanding of different software engineering principles, my problem-solving skills have dramatically improved, and my analytical skills are the best they've ever been. After learning everything I've learned I am looking forward to continuing my professional development after I graduate and applying my skills in real-world environments, and even tackling bigger, more challenging projects. This ePortfolio is a representation of the culmination of my entire Computer Science degree, it showcases all of the technical and professional growth I've had over the last few years. These artifacts help show this progression from basic assignments meant to get a good grade and nothing more, to professional-level solutions that could be implemented into real-world solutions.